

Software Requirements Specification

for

cs3337-Project

Version 2.0 approved

**Prepared by:
Jorge Arias, David Lopez, Luis Rosas, Weston Wood, Jordy Ye Cao,**

Sponsor: California State University, Los Angeles

Date: May 8, 2024

Table of Contents

Table of Contents.....	pg 2
Revision History.....	pg 3
1. Introduction.....	pg 4
1.1. Purpose.....	pg 4
1.2. Intended Audience and Reading Suggestions.....	pg 4
1.3. Product Scope.....	pg 4
1.4. Definitions, Acronyms, and Abbreviations	pg 4
1.5. References.....	pg 4
2. Overall Description.....	pg 5
2.1. System Analysis.....	pg 5
2.2. Product Perspective.....	pg 5
2.3. Product Functions.....	pg 5
2.4. User Classes and Characteristics.....	pg 6
2.5. Operating Environment.....	pg 6
2.6. Design and Implementation Constraints.....	pg 6
2.7. User Documentation.....	pg 7
2.8. Assumptions and Dependencies.....	pg 7
2.9. Apportioning of Requirements.....	pg 7
3. External Interface Requirements.....	pg 8
3.1. User Interfaces.....	pg 8
3.2. Hardware Interfaces.....	pg 8
3.3. Software Interfaces.....	pg 8
3.4. Communications Interfaces.....	pg 8
4. Requirements Specification.....	pg 9
4.1. Functional Requirements.....	pg 12
4.2. External Interface Requirements.....	pg 12
4.3. Logical Database Requirements.....	pg 12
4.4. Design Constraints.....	pg 14
5. Other Nonfunctional Requirements.....	pg 15
5.1. Performance Requirements.....	pg 15
5.2. Safety Requirements.....	pg 15
5.3. Security Requirements.....	pg 15
5.4. Software Quality Attributes.....	pg 15
5.5. Business Rules.....	pg 15
6. Other Requirements.....	pg N/A
Appendix A: Glossary.....	pg 17
Appendix B: Analysis Models.....	pg 18
Appendix C: To Be Determined List.....	pg 19

Revision History

Name	Date	Reason For Changes	Version
Weston Wood	4/23/2024	Base UI Overhaul	Draft 1
Jordy Ye Cao	04/23/2024	Added a comments feature	Draft 1
David Lopez	04/24/2024	Added search bar feature	Draft 2
Jorge Arias	4/27/2024	Added a shopping cart feature	Draft 2
Luis Rosas	4/30/2024	Favorites feature	Draft 2
Jorge Arias	5/2/2024	Added genres & changed the related files	Draft 3
Jorge Arias	5/2/2024	Updated Shopping cart	Draft 3
Weston Wood	5/2/2024	Updated UI for multiple pages	Draft 3
Weston Wood	5/2/2024	Added About Us page	Draft 3

1. Introduction

1.1 Purpose

It serves as a roadmap that outlines the features expected in the identified version, including the system's objectives and functionalities. It guides the development process by clearly defining what the software is supposed to achieve without delving into implementation specifics.

1.2 Intended Audience and Reading Suggestions

The professor of CS 3337 is our audience, and this document is meant to help the professor understand the features & changes that have occurred due to our group cooperation.

1.3 Product Scope

The Web Application works as a “Book Store” that allows the user to purchase & sell books, see the details of books, and provide an option to search for a book.

- Purchasing is done through a shopping cart feature that allows the user to add the desired book to a list showing the total price they will need to pay.
- Seeing the details of the book allows the user to view the reviews/comments that other users have created. This helps the user see if the book is to their liking.

1.4 Definitions, Acronyms, and Abbreviations

If any, Refer to Appendix A

1.5 References

- Software Requirement Template
Author: Jiang Guo
Date: April 18, 2024

2. Overall Description

2.1 System Analysis

Our website endeavors to recreate the immersive experience of browsing through a physical library, fostering user engagement and satisfaction. Our primary goal is to establish an intuitive platform where users can effortlessly discover and access a wide array of books. To achieve this, we prioritize creating a visually appealing interface, ensuring seamless navigation and interaction.

2.2 Product Perspective

This exchange website operates as an independent platform that simulates the functions that can occur in a library. It provides users with functionalities for searching, selling, and buying books within a virtual library environment.

While similar systems may exist in the market, this software distinguishes itself through its user-friendly interface, search capabilities, and seamless integration of selling and buying functionalities. Unlike some existing platforms that may focus solely on book sales or library cataloging, our software combines these features into a cohesive and intuitive user experience. Additionally, the motivation behind creating this software lies in providing users with a convenient and efficient platform to explore, trade, and acquire books, catering to both sellers and buyers in a single environment.

2.3 Product Functions

2.3.1 Homepage: function as an index page for users

2.3.2 Registration: Create an account for users

2.3.3 Login/Logout: Sign in or out of account.

2.3.4 Post Book: Allow users to upload their own books.

2.3.5 Display Books: Show all books in the website database

2.3.6 Book Details: Display detailed book information.

2.3.7 My Favorites: Allow users to have a personal favorites list of books.

2.3.8 Shopping cart: Allow users to add or remove books that they want to purchase.

2.3.9 Checkout Function: User enters details to complete the purchase of books

2.3.10 Comments Function: Allow users to write comments on books from Book Details.

2.3.11 My Books: The user can view all books that they have posted to the website.

2.3.12 Search Book: Users can search for books that have matching keywords.

2.3.13 About Us: Display information regarding the project.

2.4 User Classes and Characteristics

Novice:

- Characteristics: Users with low to intermediate knowledge of websites.
- Usage: Occasional or infrequent usage.
- Features available: Basic browsing and book search functionalities.
- Technical Expertise: Limited technical proficiency.
- Educational Level/Experience: Varied educational backgrounds and experience levels.
- Importance: This user class is crucial as they represent potential new users of the product. Ensuring a simple UI is important in facilitating their engagement with the platform.

Expert:

- Characteristics: Users with intermediate to advanced knowledge of websites.
- Frequency of Use: Regular or frequent usage.
- Usage: Utilize advanced features such as selling, buying, and managing book listings.
- Technical Expertise: Moderate to high technical proficiency.
- Educational Level/Experience: Varied educational backgrounds and extensive experience with online platforms.
- Importance: While this user class may represent a smaller portion of the user base, their engagement is vital for the platform's success. Providing comprehensive features and smooth usability for experienced users enhances the overall user experience and promotes platform loyalty.

2.5 Operating Environment

The book exchange can function fine as long as the device can access the internet and has a web browser.

2.6 Design and Implementation Constraints

2.6.1 Web framework: Django web framework

2.6.2 Python: building blocks

2.6.3 Bootstrap: visuals

2.7 User Documentation

User documentation will not produced for the website.

2.8 Assumptions and Dependencies

2.8.1 Assumptions

2.8.1.1 The user's device screen size should be sufficient to display the website's content.

2.8.1.2 The user has a browser that is compatible with the features of the website.

2.8.1.3 The user has basic knowledge of how to navigate and interact with the website.

2.8.2 Dependencies

2.8.2.1 User interface dependent on Bootstrap

2.9 Apportioning of Requirements

Currently, no requirements are set to be included in future versions of the systems.

3. External Interface Requirements

3.1 User Interfaces

The software product encompasses two distinct UIs, each designed to cater to different user states and needs.

- **Logged-Out Interface:** This interface is accessible to users who have not yet logged into the system. It presents essential elements such as the book display, search bar, and login functionality. Users can browse available books and initiate the login process to access additional features.
- **Logged-In Interface:** Upon successful authentication, users gain access to an expanded set of features. In addition to the functionalities available in the logged-out interface, users can now interact with the shopping cart, post books, and view their uploaded books. This interface provides a comprehensive and personalized experience, allowing users to manage their interactions with the platform effectively.

3.2 Hardware Interfaces

The project is accessible without any required hardware interfaces.

3.3 Software Interfaces

As of now: Since the project exclusively relies on the Django admin system, there are no additional software products or interfaces required. The Django admin system itself provides the necessary tools to fulfill the project requirements.

Similarly, there is no need to interact with external APIs as all functionalities are contained within the Django admin system.

3.4 Communications Interfaces

3.4.1 The website supports web browsers, using HTTP to ensure secure communication between the user and the database.

3.4.2 Registration form to make an account and password

4. Requirements Specification

4.1 Functional Requirements

4.1.1 Application Requirements

4.1.1.1 The system shall run on Windows, Mac, and Linux.

4.1.1.2 The system shall provide a login interface for users to input their credentials and prompt users to re-enter their credentials if initial input is incorrect

4.1.1.3 The application shall provide an interface for the following functions:

4.1.1.3.1 The system shall provide a Homepage

4.1.1.3.2 The system shall provide a Registration function.

4.1.1.3.3 The system shall provide a Login/Logout function.

4.1.1.3.4 The system shall provide a Post Book function.

4.1.1.3.5 The system shall provide a Display Books function.

4.1.1.3.6 The system shall provide a Book Details function.

4.1.1.3.7 The system shall provide a My Favorites function.

4.1.1.3.8 The system shall provide a Shopping Cart function.

4.1.1.3.9 The system shall provide a Checkout function.

4.1.1.3.10 The system shall provide a Comments function.

4.1.1.3.11 The system shall provide a My Books function.

4.1.1.3.12 The system shall provide a Search function.

4.1.1.3.13 The system shall provide an About Us function.

4.1.2 Home Page

4.1.2.1 The homepage shall serve as the starting landing point for users visiting the website.

4.1.2.2 Website navigation will be limited to public users, while logged-in users shall have access to all website functions.

4.1.2.3 The home page shall display books from a featured genre.

4.1.3 Registration Function

4.1.3.1 The registration function shall allow users to create an account for the website using Django's registration form.

4.1.4 Login/Logout Function

4.1.4.1 The login function shall allow users to access their account and, upon valid input, redirect the user back to the homepage.

4.1.4.2 Logging in shall grant users access to website functions.

4.1.4.3 The logout option shall be present at the top right corner and redirect the user back to the homepage.

4.1.5 Post Books Function

4.1.5.1 The Post Book function shall display a form generated from Django.

4.1.5.2 The application shall ask the user the following inputs:

4.1.5.2.1 Book Name

4.1.5.2.2 Book URL

4.1.5.2.3 Book price

4.1.5.2.4 Book image

4.1.5.3 When the form is submitted, the application shall add the book to the website's database, including the username of the submitter and the time of submission.

4.1.6 Display Books Function

4.1.6.1 The Display Books function shall display all books in the website's database.

4.1.6.2 The display contents shall include the following:

4.1.6.2.1 The system shall display the name of the book.

4.1.6.2.2 The system shall display the book image.

4.1.6.2.3 The system shall display book genres.

4.1.6.2.4 The system shall include the option to add comments.

4.1.6.2.5 The system shall include the option to view comments.

4.1.7 Book Details Function

4.1.7.1 Book Details shall display the selected book details.

4.1.7.1.1 The system shall display the book name.

4.1.7.1.2 The system shall display books genre(s).

4.1.7.1.3 The system shall display the price of the book.

4.1.7.1.4 The system shall display the username of the submitter.

4.1.7.1.5 The system shall display the date the book was submitted.

4.1.7.1.6 The system shall display the following:

4.1.7.1.6.1 "Add to Cart" add the selected book to the user's shopping cart.

4.1.7.1.6.2 "Add to Favorites" add the selected book to the user's favorites list.

4.1.8 My Favorites Function

4.1.8.1 From the Book Detail page, the system shall display:

4.1.8.1.1 "Add to Favorites" button if the book already exists in the favorites list then it should not duplicate the book entry.

4.1.9 Shopping Cart Function

4.1.9.1 Shopping Cart function shall be available for users who are logged in.

4.1.9.2 Shopping Cart shall display the books users have added to their cart.

4.1.9.3 Shopping Cart page shall display the following:

- 4.1.9.3.1 Title of each book
- 4.1.9.3.2 Price of each book
- 4.1.9.3.3 Quantity of each book
- 4.1.9.3.4 Cart summary: cost of all books in cart
- 4.1.9.3.5 Link that removes the book from cart
- 4.1.9.3.6 Link that allows user to include a gift package to order
- 4.1.9.3.7 Checkout button
- 4.1.9.3.8 Cancel button, that removes all books in the cart and redirects the user to the homepage.

4.1.9.4 Selecting the checkout button redirects the user to the checkout page.

4.1.10 Checkout Function

- 4.1.10.1 The checkout function shall prompt a form requesting the following:
 - 4.1.10.1.1 Name
 - 4.1.10.1.1 Email
 - 4.1.10.1.1 Address
 - 4.1.10.1.1 Payment method(Credit Card/PayPal)
- 4.1.10.2 Placing order shall require all input fields to be filled out.
- 4.1.10.3 The system shall display the Order summary.
- 4.1.10.4 When an order is submitted, the application shall redirect the user to a page informing them that the transaction was successful.

4.1.11 Comments Function

- 4.1.11.1 Comment Function shall allow the user to add comments to a book.
 - 4.1.10.1.1 The system shall prompt the user to input text for the comment.
- 4.1.11.2 Comment Function shall allow the user to view all comments of a book

4.1.12 My Books Function

- 4.1.12.1 The My Books function shall allow users to view all books they have submitted.
- 4.1.12.2 The My Books function shall show the following book details:
 - 4.1.10.1.1 Book name
 - 4.1.10.1.2 Book genre(s)
 - 4.1.10.1.3 Book image
- 4.1.12.3 A link shall be present for each book, redirecting to the Book Detail page.
- 4.1.12.4 A button shall be provided to remove a book submission from the user.

4.1.13 Search Function

4.1.13.1 The Search Function shall present a search bar in the center of the navigation bar so that the user can search for a book.

4.1.13.2 When using the search function, that website database will be scanned to match keywords with book entries.

4.1.13.3 The Search function shall redirect the user to a results page and display:

4.1.13.3.1 The book names that match the search keyword and will link to its details page.

4.1.13.3.2 Book image

4.1.13.4 If no matches are found, the Search Function shall redirect the user to a page, informing them that no results were found.

4.1.14 About Us Function

4.1.14.1 The About Us Function shall inform users about the website's purpose.

4.1.14.2 The About Us functions shall also feature details regarding the contributors involved in developing the website.

4.2 External Interface Requirements

4.2.1 User Interfaces

4.2.1.1 Website interface built using HTML and CSS framework Bootstrap.

4.2.1.2 Hardware Interfaces

No additional hardware interfaces are needed for the system.

4.2.1.3 Software Interfaces

No additional software interfaces are needed for the system.

4.2.1.4 Communication Interfaces

The system uses a client-server model, use HTTPS methods

4.2.1.4.1 GET Method, used to access account and unlock website functions.

4.2.1.4.1 POST Method, used to protect billing information.

4.3 Logical Database Requirements

4.3.1 Database Information

4.3.1.1 User Data

4.3.1.1.1 Username

4.3.1.1.2 Password

4.3.1.1.3 Account Type(Admin, Registserd, Guest)

4.3.1.2 Book Data

4.3.1.2.1 Book name

4.3.1.2.2 URL

- 4.3.1.2.3 Price
 - 4.3.1.2.4 Publish Date
 - 4.3.1.2.5 Genre(s)
 - 4.3.1.2.6 Image
 - 4.3.1.2.7 Username of uploader
 - 4.3.1.3 **My Books Data**
 - 4.3.1.3.1 Identification number for user
 - 4.3.1.3.2 Book entry ID
 - 4.3.1.4 **User Comments**
 - 4.3.1.4.1 Username
 - 4.3.1.4.2 Comment text
 - 4.3.1.4.3 Book ID number
 - 4.3.1.4.4 Comment ID number
 - 4.3.1.5 **Shopping Cart**
 - 4.3.1.5.1 User account ID number
 - 4.3.1.5.2 Book ID number
 - 4.3.1.6 **Favorites List**
 - 4.3.1.6.1 Book ID number
 - 4.3.1.6.2 Many-to-Many relationship between user and book data.
- 4.3.2 **Frequency of Use**
 - 4.3.2.1 **Book Data**
 - 4.3.2.1.1 Used to display data of books to users
 - 4.3.2.2 **User Data**
 - 4.3.2.2.1 Access account
 - 4.3.2.2.2 Browse book content, upload book, and purchase books
- 4.3.3 **Accessing Capabilities**
 - 4.3.3.1 **HTTP Methods**
 - 4.3.3.1.1 GET request book data to display to users.
 - 4.3.3.1.2 POST sends user data to the website database and makes updates.
 - 4.3.3.2 **User and Book Data**
 - 4.3.3.2.1 User and Book data should only be accessed and modified by functions pertaining to their data.
 - 4.3.3.2.2 Access privileges to User and Book data assigned to account types: admin and moderators.

4.3.4 Data Entities and their Relationships

4.3.4.1 To access the website's book content and functions, users must have a registered account and log in to be granted access.

4.4 Design Constraints

4.4.1 U.I. Responsive to all different devices

4.4.1.1 Scale proportional to device screen size

4.4.1.2 Interaction, mobile touch input, and desktop mouse and keyboard input

4.4.1.3 Device capabilities and performance

4.4.2 Use of Bootstrap

CSS framework used to create the interface of the website

4.4.3 Use of Django

Web framework that uses model–template–views architectural pattern to build the website.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The website operates using Django 3.11, which supports simultaneous users and requests. Users with stable connections should expect transactions and requests to be processed within a second. Website page navigation, hyperlinks, and buttons should be seamless, and the use of functions should not be delayed.

5.2 Safety Requirements

Regarding user account information, users should keep their login details private. It is recommended to follow password practices such as having 12 or more characters, having special characters or numbers, avoiding including personal information in passwords, and regularly updating account password. Furthermore, if any concerns about account details and payment methods arise, feel free to contact customer support.

5.3 Security Requirements

For the website to be completely accessible, users should have a registered account and be logged in. Being a registered user ensures that any personal information associated with the account is protected and will not be disclosed to any outside parties. The website also uses secure communication methods, such as HTTPS, to carry out communication across different functions of the system.

5.4 Software Quality Attributes

The website is functional in different browsers across different devices. The structure of the system makes it easily maintainable and supports modifications and updates in the future. The system structure also ensures that it is scalable to handle increases in resources and traffic. Additionally, the system focuses on sustainability, optimizing functions, and not over-consuming energy.

5.5 Business Rules

Users who are registered and logged in shall have website privileges to view book details, upload and sell books, purchase books, and make comments. All transactions regarding selling and purchasing books are in compliance with the website's terms and services agreement.

6. Legal and Ethical Considerations

The following is regarding any legal and ethical issues associated with this project.

6.1 Product Listings

6.1.1 Accurate and detailed information about books.

6.1.2 No prohibited material.

6.2 Intellectual Property Rights

6.2.1 No sale of counterfeit books.

6.2.2 Sale of authentic copies of books.

6.3 Data Protection

6.3.1 User data in compliance with regulations regarding E-commerce.

6.4 User Conduct

6.4.1 Fair pricing: users shall not engage in price gouging and other related pricing methods that exploit other users.

6.4.2 Transparency on the item being sold is as listed.

6.4.3 No malicious/harmful activities, including but not limited to:

6.4.3.1 Fraud

6.4.3.2 Scamming

6.4.3.3 Harassment

6.4.3.4 Inaccurate information

6.4.3.5 Spamming of posts or comments

6.4.3.6 Offensive behavior towards groups

6.5 Shipping and Delivery

6.5.1 Follow shipping regulations and standards.

6.5.2 Proper Shipping labels and documentation.

6.5.3 Packing should be done correctly to reduce the risk of damage.

6.6 Accessibility

6.6.1 Enhanced user experience for all users

6.6.1.1 Ease of access to website functions

6.6.1.2 Text readability

6.6.1.3 Continuous improvement and maintenance of the website.

Appendix A: Glossary

Django: a python-web-based framework that uses model-template-views architectural pattern.

U.I: User Interface

HTTPS: Hypertext Transfer Protocol Secure, which is used to secure communication and data transfer between the user's web browser and website.

HTML: Hyper Text Markup Language.

CSS: Cascading Style Sheets

Bootstrap: CSS framework

ID: Identification

Appendix B: Analysis Models

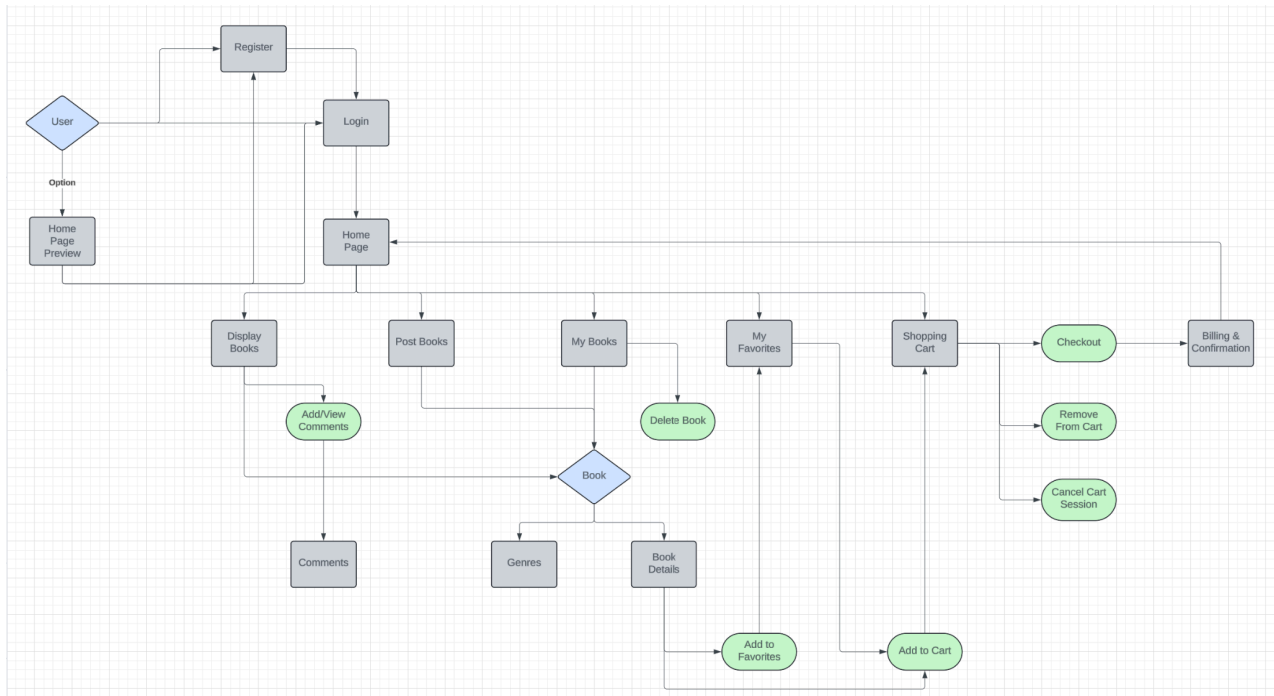


Figure 1. System Structure of Functions.

Appendix C: To Be Implemented List

- Account Details
- Book Rating
- Purchase History
- Comment History
- Recommendations
- Community Forms